Week 6 Problem Set: The Great Theorems

Calculus Honors · Module 3: Continuity & the Great Theorems · Lessons 3.03 and 3.04

Name: _____ Date: _____

> Show all of your work. A correct answer with no reasoning earns partial credit at most. Every IVT or EVT argument must name the hypotheses you checked, in full sentences, before stating the conclusion.

PART A · PROVING A ROOT EXISTS

1. Prove that $f(x) = x^3 - 4x + 1$ has a root on $[0, 1]$. Follow all four steps: continuity, endpoint values, sign change, conclusion.

2. Prove that $g(x) = \sqrt{x} - x + 1$ has a root on $[2, 4]$. State all four steps.

PART B · THE INTERMEDIATE VALUE THEOREM, MORE GENERALLY

3. Let $f(x) = x^2$, continuous on $[1, 4]$. Does the Intermediate Value Theorem guarantee some c in $(1, 4)$ with $f(c) = 9$? Justify using the theorem's statement, not just by solving for c directly.

---

4. A function h has an infinite discontinuity at $x = 3$, and is defined on $[1, 5]$ with $h(1) = -2$ and $h(5) = 6$. Does the Intermediate Value Theorem guarantee a value of x where $h(x) = 0$? Explain which hypothesis is or is not met.

PART C · THE EXTREME VALUE THEOREM AND ITS HYPOTHESES

5. Does the Extreme Value Theorem guarantee an absolute maximum and minimum for $f(x) = x^2 - 2x$ on $[0, 3]$? Justify by checking both hypotheses.

---

6. Let $g(x) = \dfrac{1}{x}$ on the open interval $(0, 2)$. Explain why the Extreme Value Theorem does not guarantee a maximum here, naming exactly which hypothesis fails.

---

7. Let $h(x) = \begin{cases} x & 0 \le x < 2 \\ 0 & x = 2 \end{cases}$ on the closed interval $[0, 2]$. Explain why the Extreme Value Theorem does not guarantee a maximum here, naming exactly which hypothesis fails.

PART D · EXPLAIN YOUR THINKING

8. Explain why an existence theorem like the IVT can be genuinely useful even though it never tells you the exact value of the root or extremum it guarantees.

9. Convince a classmate, using one of your counterexamples from Part C, why checking both hypotheses of a theorem matters before applying its conclusion.